

# Algebra II Practice: Parabolas and Distributive Law

## Instructions

Complete in 30 minutes. Show all work. Give exact answers unless a decimal is explicitly requested.

## Problems

1. Expand and simplify:

$$3(2x - 5)^2 - 4(x + 1)(x - 6).$$

2. Expand and simplify:

$$(x - 3)(2x + 5)(x + 4).$$

3. Factor completely:

$$4x^3 - 16x^2 - 9x + 36.$$

4. Factor completely:

$$2x^4 - 13x^2 - 7.$$

5. Solve:

$$5(2x - 1)^2 - 3(x + 4) = 2x^2 + 17.$$

6. Solve:

$$(x - 2)(x + 5) = 3(2x - 1).$$

7. For  $f(x) = 2x^2 - 12x + 5$ , find the vertex, axis of symmetry, and minimum value.

8. Write  $g(x) = -3x^2 + 18x - 20$  in vertex form, then state the maximum value.

9. For  $h(x) = x^2 - 4x - k$ , find the value(s) of  $k$  so the parabola has exactly one  $x$ -intercept.

10. A parabola has  $x$ -intercepts  $-2$  and  $6$  and passes through  $(1, -18)$ . Write its equation in standard form.

11. A rectangle has perimeter 40 units. Write its area as a quadratic function of its width, then find the maximum possible area.
12. A ball's height is modeled by  $H(t) = -16t^2 + 72t + 4$ . Find the maximum height and the time when it occurs.